

Gradient Descent

Effect of Learning Rate

GD Example:

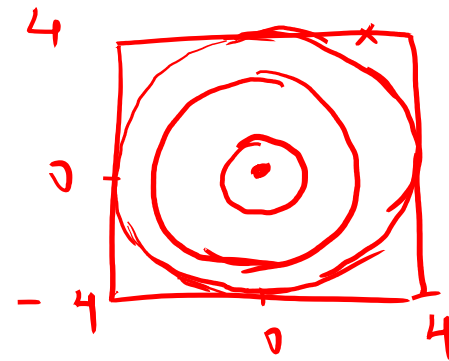
$$J(\theta) = \theta_1^2 + \theta_2^2 + 4$$

$$\frac{\partial J(\theta)}{\partial \theta_1} = 2\theta_1$$

$$\frac{\partial J(\theta)}{\partial \theta_2} = 2\theta_2$$

$$\theta_1^{(k+1)} = \theta_1^{(k)} - \alpha (2\theta_1^{(k)})$$

$$\theta_2^{(k+1)} = \theta_2^{(k)} - \alpha (2\theta_2^{(k)})$$



We know the global minima is at $(0, 0)$

⇒ Initialize

$$\theta_1 = 3$$

$$\theta_2 = 4$$

$$\alpha = 2, 1, 0.1, 0.5$$

① $\alpha = 2$.

Iteration (k)	$\vec{\theta}$	$\nabla J(\vec{\theta})$	J	$\theta^{k+1} = \theta^k - \alpha \nabla J$
0	(3, 4)	(6, 8)	29	$3 - 2 \times 6 = -9$ $4 - 2 \times 8 = -12$
2	(-9, -12)	(-18, -24)	229	$-9 - 2 \times (-18) = 27$ $-12 - 2 \times (-24) = 36$
3	(27, 36)	(54, 72)	2029	$27 - 2 \times (54) = -81$ $36 - 2 \times (72) = -108$

(Cost is increasing)

② $\alpha = 1$.

k	θ	$\nabla J(\theta)$	γ	θ^{k+1}
0	$(3, 4)$	$(6, 8)$	29	$3 - 1 \times 6 = -3$ $4 - 1 \times 8 = 0 -4$
1	$(-3, -4)$	$(-6, -8)$	29	$-3 - 1 \times (-6) = 3$ $-4 - 1 \times (-8) = 4$
2	$(3, 4)$	$(6, 8)$	29	$(-3, -4)$

Oscillating without Divergence or Convergence.

③ $\alpha = 0.1 \rightarrow$ Slow Convergence

k	θ	$\nabla J(\theta)$	J	θ^{k-1}
0	(3, 4)	(6, 8)	29	(2.4, 3.2)
1	(2.4, 3.2)	(4.8, 6.4)	20	1.92, 2.56
2	1.92 2.56	3.84, 5.12	14.24	1.54, 2.05
...	...	...	...	...
30	0.0037 0.005	...	4	...

④ $\alpha = 0.5$

k	θ	∇J	J	θ^{k-1}
0	3, 4	6, 8	29	$3 - 0.5 \times 6 = 0$ $4 - 0.5 \times 8 = 0$

Stopping Criteria: -

$$(i) \quad \|\vec{\theta}^{(k+1)} - \vec{\theta}^{(k)}\| \leq \epsilon$$

$$(ii) \quad \|\nabla_{\theta} J(\theta^{(k)})\| \leq \epsilon$$

$$(iii) \quad |J(\theta^{(k+1)}) - J(\theta^{(k)})| \leq \epsilon$$

$$\vec{x} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

$$\|\vec{x}\| = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{4 + 9 + 16} = \sqrt{29}$$

* Claim:- The direction of maximum rate of change for a
fn. $J(\theta)$ is given by $\nabla_{\theta} J(\theta)$

Proof:- Let $\vec{v}$ is a unit vector.

The rate of change in a given direction ($\vec{v}$) is $\frac{\partial J}{\partial \vec{v}}$

$$\boxed{\frac{\partial J}{\partial \vec{v}}} = \nabla J \cdot \vec{v}$$
$$= \vec{G} \cdot \vec{v}$$

$$\boxed{\vec{G} = \nabla J}$$

$$= |\vec{G}| |\vec{v}| \underline{\cos \theta}$$